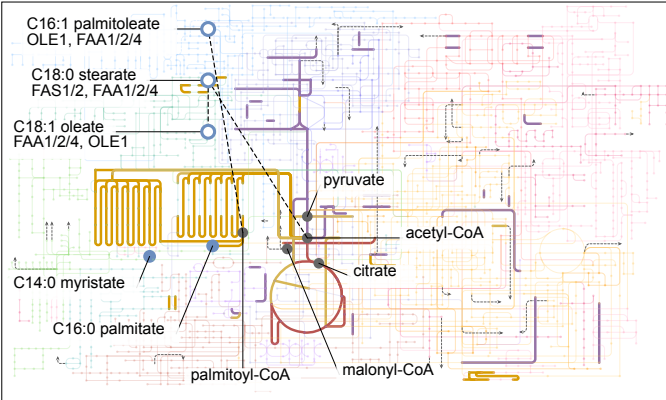




the route from central carbon to the measured lipids

- glycolysis and gluconeogenesis
 - pyruvate metabolism
 - citrate cycle
 - fatty acid biosynthesis, elongation and desaturation
 - fatty acid degradation
 - measured species, a node of the reference map
 - measured species attached from the yeast GEM
 - named intermediate
 - the GEM's shortest path from a drawn compound; its genes under the name
- Palmitoleate, stearate and oleate have no node on the reference map, as acids or acyl-CoAs; each is attached along the yeast GEM's shortest path from a drawn compound. Of the 10 deleted regulators, 0 draw a reaction on this map; of the 13 pathway genes of b, 9 do and ELO1/2/3, OLE1 do not.